

Analysis of Crop Grown in India

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Abstract

This project focuses on analysing and visualising crop production and yield patterns in India over the past 45 years on specific and most-grown crops, including Wheat, Rice, Maize, Cotton, and Sugarcane, aiming to provide a clear idea for farmers and researchers.

The study involves the filtration and cleaning of the dataset which includes crop production, yield, and other relevant parameters. Utilising Power BI, the dataset is converted into interactive visualisations, particularly emphasizing the relationship between crop production and crop yield. These visualizations, presented in the form of graphs and charts, give a clear image of the trends and patterns observed over the years from 1970-2017. The primary objective is to enable farmers to analyse the performance of crops in their respective states and districts, to ease decision-making regarding crop selection and agricultural practices in their respective regions.

Additionally, the research benefits researchers in the study of crop cultivation on the environment. By using advanced data analytics and visualization techniques, this research helps in agricultural productivity and sustainability in India. The analytics from this study play an important role in crop management, lowering the risks linked to variations in production and yield, and promoting eco-friendly farming methods. This research report also acts as a significant tool for stakeholders, policymakers, and researchers, allowing them to make decisions that contribute to the progress of agricultural and environmental sustainability in India.

Introduction

Understanding the dynamics of crop cultivation, especially for major crops like Wheat, Rice, Maize, Cotton, and Sugarcane, is crucial. The availability of detailed datasets allows for in-depth analysis, offering valuable insights for informed decision-making and sustainable agricultural practices.

This study focuses on analyzing crop cultivation patterns in India from 1970 to 2017. The dataset includes key variables such as state code, district code, crop area, production, yield, and year of data, enabling a detailed examination at both state and district levels. By utilizing Power BI for data cleaning and filtering, the dataset is prepared for informative visualizations.

Interactive graphs and visual representations aim to highlight production, yield, and cultivation area trends of the major crops across different states and districts in India. Wheat, Rice, Maize, Cotton, and Sugarcane are the main focus, allowing for a comparative analysis of their performance over the years. This comparison reveals insights into regional variations and trends, shedding light on the strengths and weaknesses of each crop.

The findings of this analysis will be shared through a user-friendly website built with HTML, ensuring accessibility for a wide range of stakeholders. By providing a comprehensive view of crop cultivation dynamics, this project seeks to empower farmers with actionable insights to enhance their agricultural practices. It also serves as a valuable resource for policymakers, researchers, and agriculture enthusiasts.

Objective

The objective is to understand and visualize the trends and patterns of the major crops grown in India, namely: Wheat, Rice, Maize, Cotton, and Sugarcane at both regional and national levels using data from the past 45 years in fields like agricultural production, identifying areas of improvement, and formulating policies for sustainable agricultural development.

To identify which crop is most preferred by farmers according to the area and region. It includes observation of the change in crop selection from 1970-2017

Design

This research paper shows importance of interactive dashboards in data visualization and analysis. It includes design framework that consists of various visuals like tree, line, column, bar, pie, doughnut, and area line. The interactive dashboard consists of six pages, with one main page and five descriptive pages, all created using Power BI. It offers a structured way to represent data and tools to navigate and interpret complex datasets effectively. This implementation includes features like zoom-in/out, hover-over tooltips, clickable elements, and dynamic filtering options to enhance user experience and facilitate data exploration. Main objective is to provide a theoretical scenario to show the dashboard's utility, its use and importance in decision-making with valuable insights derived from thorough data analysis.

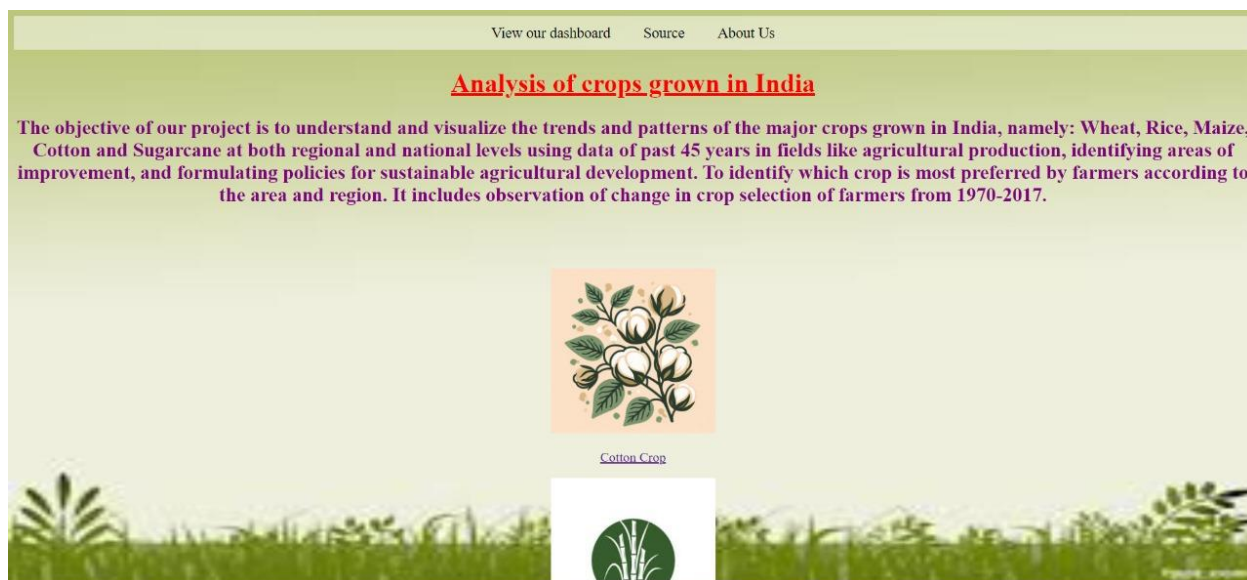


Fig. 1.1

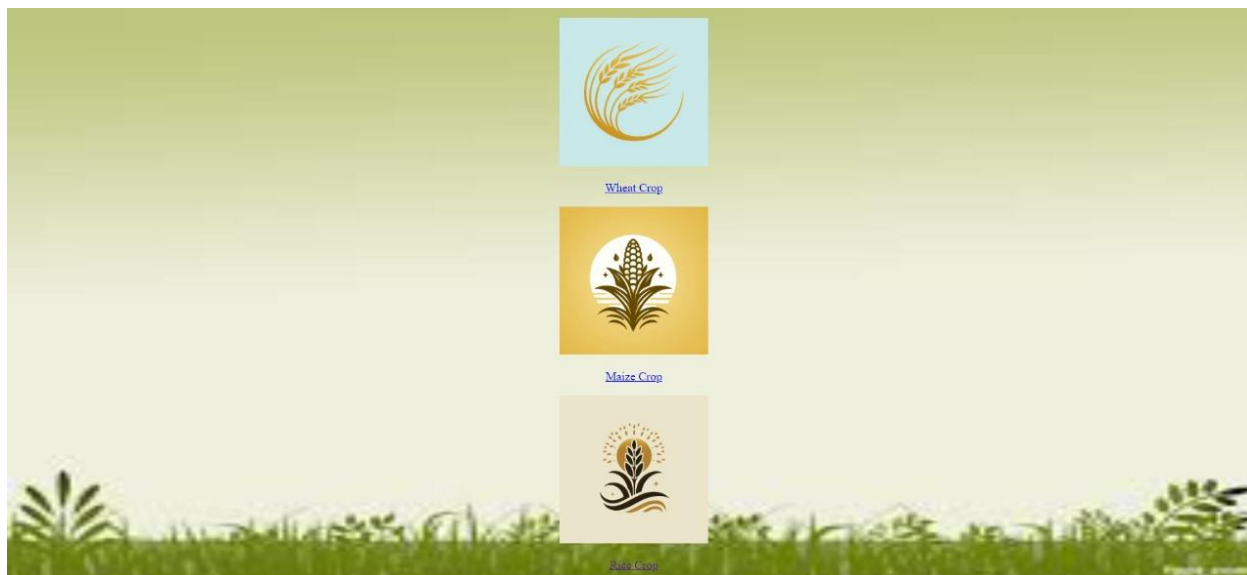


Fig1.2

Fig 1.1 and Fig 1.2 is website that is created using HTML and CSS to provide a basic platform to users which allows them to view dashboard. Additionally, there are some pictures added to view dashboard of specific crops and analyse them. The 'Source' tab was particularly helpful as it provided access to dataset we used to create dashboard and visualisations.

Visualization

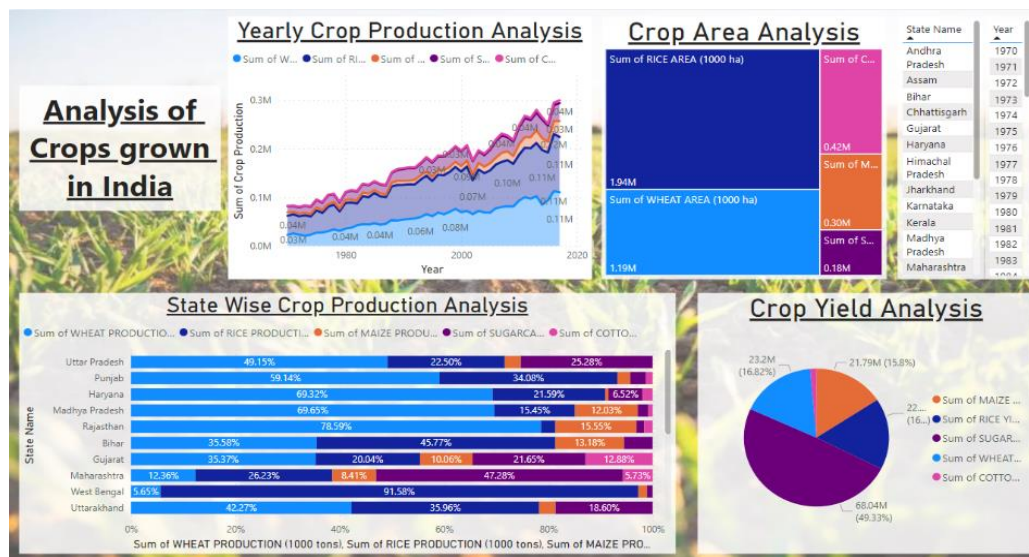


Fig. 1 - Dashboard

Fig. 1 is a compilation of graphs and charts providing an analysis of various crops grown in India.

- Yearly Crop Production Analysis: The chart shows the sum of wheat, rice, maize, sugarcane, and cotton production over the years.
- Crop Area Analysis: This bar graph displays the sum of rice and wheat area in different states of India.
- State-Wise Crop Production Analysis: This horizontal bar graph shows the sum of wheat, rice, maize, sugarcane, and cotton production in various states.
- Crop Yield Analysis: This pie chart depicts the yield proportions of maize, rice, sugarcane, wheat, and cotton.

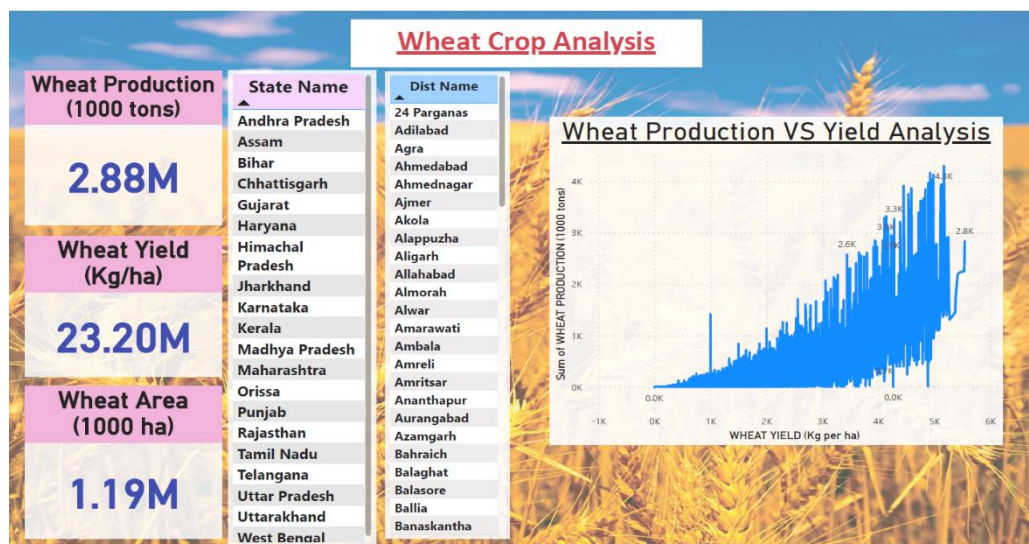


Fig. 2 – Wheat Crop Analysis

Fig. 2 is a detailed graphical representation of wheat crop analysis in India. It includes:

- Wheat Production: The image shows data on wheat production in India, with a total production of 2.88 million tons.
- Wheat Yield: The yield of wheat is represented as 23.20 million kg per hectare.
- Wheat Area: The area covered by wheat crops is 1.19 million hectares.
- “Wheat Production VS Yield Analysis,” illustrates the correlation between production and yield, indicating that as the yield increases, so does the production.

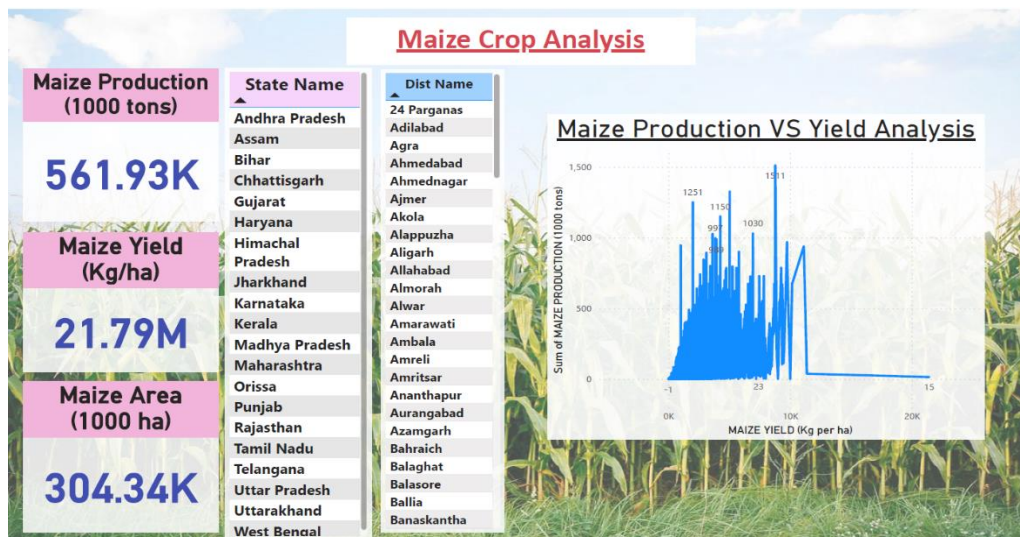


Fig. 3 – Maize Crop Analysis

Fig. 3 is an infographic that presents data on maize production in India. It includes:

- Maize Production: The total production is listed as 561.93K tons.
- Maize Yield: The yield of maize is noted as 21.79M Kg per hectare.
- Maize Area: The area covered by maize crops is 304.34K hectares.
- “Maize Production VS Yield Analysis,” compares the production and yield of maize, indicating that higher yields lead to greater production.

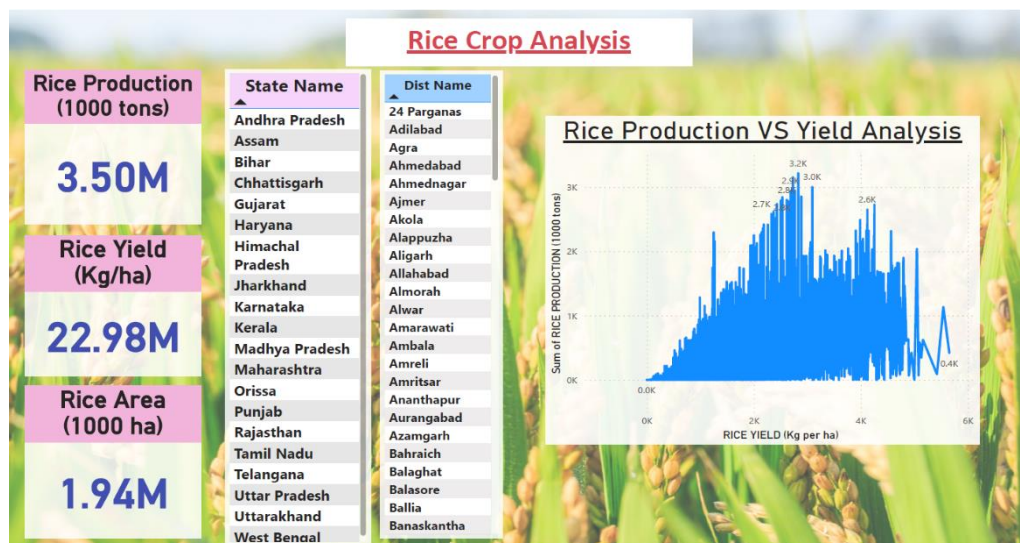


Fig. 4 – Rice Crop Analysis

Fig. 4 is an infographic on rice crop analysis in India. It features:

- Rice Production: Listed as 3.50 million tons.
- Rice Yield: Indicated as 22.98 million kg per hectare.
- Rice Area: Presented as 1.94 million hectares.
- “Rice Production VS Yield Analysis,” comparing the production and yield of rice across various states and districts in India. The background imagery of ripe rice plants adds an agricultural theme to the presentation.

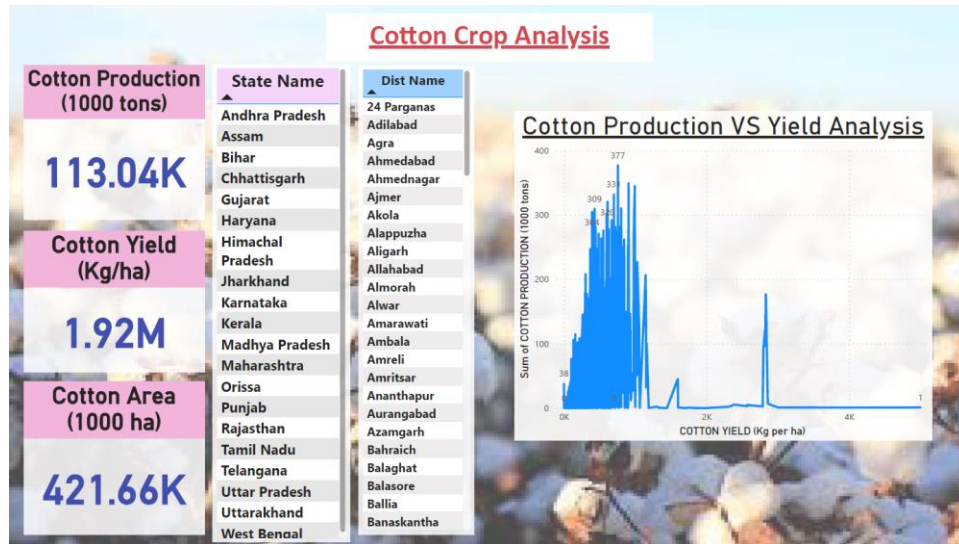


Fig. 5 – Cotton Crop Analysis

Fig. 5 is an infographic that presents data on cotton production in India.

- Cotton Production: The total production is listed as 113.04K tons.
- Cotton Yield: The cotton yield is 1.92M Kg per hectare.
- Cotton Area: The area covered by cotton crops is 421.66K hectares.
- “Cotton Production VS Yield Analysis,” compares the production and yield of cotton, indicating that higher yields lead to greater production.

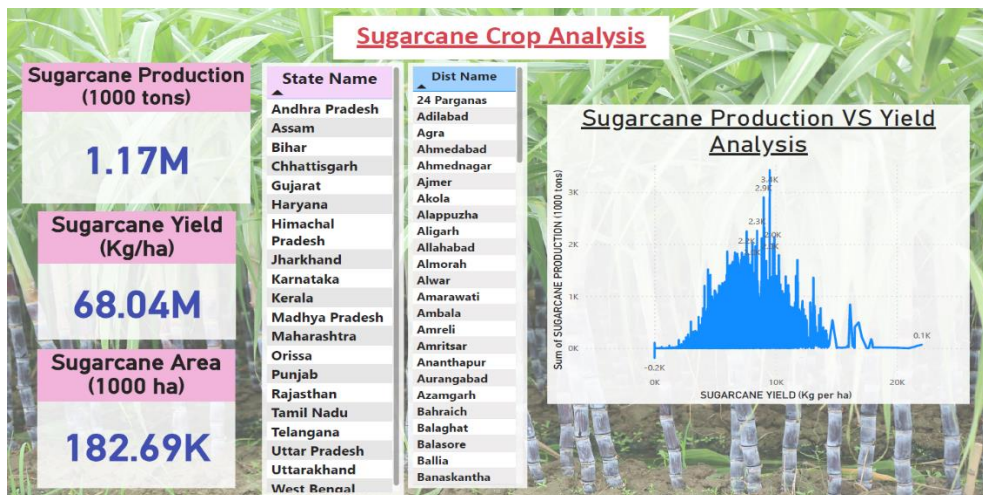


Fig. 6 – Sugarcane Crop Analysis

Fig. 6 is an infographic that provides a comprehensive analysis of sugarcane crop production in India.

- Sugarcane Production: The total production is noted as 1.17 million tons.
- Sugarcane Yield: The yield is indicated as 68.04 million kg per hectare.
- Sugarcane Area: The area covered by sugarcane crops is 182.69 thousand hectares.
- “Sugarcane Production VS Yield Analysis” compares the production and yield of sugarcane, visually representing the relationship between these two factors.

Conclusion

Based on these data, sugarcane is the most beneficial crop for achieving the highest yield. This conclusion is drawn from the analysis of crop production, yield, and area for different states. Sugarcane stands out with a high yield of 68.04M Kg/ha, which is the highest among the crops analysed, indicating that cultivating sugarcane could result in the most significant yield compared to wheat, maize, rice, and cotton.

In year 1974, there is noticeable yield of 2933.33 kg/hector of sugarcane has marked in state Haryana in two districts namely: Mahendragarh and Narnaul.

There is significant yield of wheat in year 1977 in Madhya Pradesh's Panna district of 764.54 kg/hector.

In 1974, in the Moradabad district of Uttar Pradesh, the yield of rice was 613.41 kg/hector, contributing to a total yield of 89.29 thousand kg/hector across all crops for that year.

References and Citations

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